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Participatory learning and action (PLA) to improve health outcomes in high income settings: Systematic Review Protocol

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Participatory learning and action (PLA) to improve health outcomes in high income settings: Systematic Review Protocol

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ABSTRACT

Introduction

Participatory learning and action (PLA) is a form of group reflection and learning with documented efficacy in low-income countries (LICs) to improve health-related outcomes. PLA represents both a learning philosophy and practical framework that could be applied to a variety of contexts. To date, PLA has not been widely implemented within high-income countries (HICs) to improve health and health-related outcomes. We aim to synthesise the literature currently available by means of systematic review, to form a foundation for future applications of PLA methodology in HICs.

Methods and analysis

Two reviewers will independently search pre-defined terms in the following electronic bibliographic databases: MEDLINE, EMBASE, CINAHL and the Cochrane Library. Search terms will encompass PLA and PDSA (Plan, Do, Study Act) projects, as well as studies using the Triple/ Quadruple Aim model. We will include randomised controlled trials that incorporate online or face-to-face components using PLA/PDSA methodology. Our data will be extracted into a standardised pre-piloted form with subsequent narrative review according to Synthesis without meta-analysis (SWiM) guidelines.

Ethics and dissemination

No ethics approval is required for this study. The results for this study will be submitted for publication in a leading peer-reviewed academic journal in this field. Additionally, a report will be produced for the funders of this review, which can be viewed for free on their website.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Novel analysis of available evidence of PLA efficacy within HICs
- Incorporation of multiple participatory research methods to accommodate for the variation in nomenclature
- Broad inclusion criteria to allow for wide heterogeneity of PLA study designs
- Potential low number of RCTs for inclusion

INTRODUCTION

The Participatory learning and action (PLA) approach is one of many methods encompassed in the field of participatory research methods, which also include Participatory Action Research (PAR) and Community-Based Participatory Research (CBPR).[1] In comparison to traditional research methods, the underlying philosophy for these methods is to empower communities to identify their own solutions for issues that affect them, rather than for external stakeholders to enforce a solution on to them.[2,3] The onus for researchers is to seek and include diverse opinions and ensure an open space to discuss these. In doing so, researchers can gain first-hand knowledge from communities, allowing them to often collaborate with marginalised members. There are multiple techniques that researchers can use to facilitate the process, including drawing community maps of local resources or landmarks, preference rankings to prioritise issues of the group or Venn diagrams to show the sphere of influence of particular individuals or organisations.[1,3,4]

PLA group cycle is an iterative process typically led by a facilitator who facilitates the participants through a four-stage cycle of identifying and prioritizing contextual issues, designing strategies to address these issues and a post-implementation evaluation. It aims to support members to take individual and collective action to solve health problems within their community.[5] Through direct engagement, PLA serves to empower communities to understand the importance of their health and the health of their families and communities, as well as the barriers and facilitators to optimising these health outcomes. Due to its demonstrated efficacy, the World Health Organisation has recommended the implementation of groups practicing PLA in rural low income country settings with reduced access to health services in the context of maternal and newborn health.[6,7]

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The current PLA model originated from multiple policies and theories from the last century. In his seminal work 'Pedagogy of the Oppressed', Paulo Freire defined that through critical evaluation, the 'oppressed' could re-evaluate and improve their context.[8] Around the late 1970s, Robert Chambers developed a more participatory research tool, 'Rapid Rural Appraisal', as an alternative to more didactic learning methods.[4] This tool later developed into the PLA cycle, but similar models have also emerged such as the Plan-Do-Study-Act (PDSA) cycle.[10] The PDSA cycle involves planning an intervention, delivering it and examining its effect before making adjustments based on feedback. As such, it has considerable overlap with the PLA model.

Within healthcare settings, PDSA cycles have been incorporated into the 'Model for Improvement' to expedite the Quality Improvement (QI) process.[11] QI is integral to the assessment of health services and delivering better-value, higher quality care for patients, especially important in a climate of rising healthcare costs and limited funding [12]. QI methodology can be applied to review a variety of outcomes, including waiting times, patient experience or post-operative complications. A popular framework for measuring healthcare outcomes is the 'Triple Aim' model.[13] Briefly, the three aims are as follows: improving patient experience of healthcare, improving health population and reducing per capita cost of healthcare. Increasingly, an additional fourth aim, to improve the wellbeing of healthcare workers, is integrated to form the 'Quadruple Aim'.[14]

Thus far, PLA cycles have been successfully incorporated in research to improve health outcomes. Prost et al found that women's groups that utilise PLA initiatives have reduced maternal and neonatal mortality in low-income settings (Nepal[15], India[16–18], Bangladesh[19,20] and Malawi[21,22]).[3] The most significant reduction in neonatal mortality (33%) and maternal mortality (49%) occurred in four of the sites that had the highest coverage of groups and participation of at least 30% of pregnant women. The underlying mechanisms are complex.[1,23] Some benefit can be attributed to effects such as behaviour change for hygiene, nutrition and infant care, as well as social support and birth planning. Additionally, the groups worked towards female empowerment and better decision-making to improve access to basic services, encourage safer delivery practices and reduce delays in reaching care. However, women's groups also provide poor, excluded women a voice by building their confidence to take decisions and to reduce the stigma in their lives, while possibly reducing maternal stress.

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PLA health initiatives in low-income countries (LICs) to date have largely focused on maternal and infant nutrition, but more recently have expanded to address both communicable[25] and non-communicable diseases[8,26]. For example, Tripathy et al set up a PLA women's group for pregnant mothers to reduce neonatal mortality in rural Jharkhand and Odisha in Eastern India.[18] For these groups, the facilitators were government-approved Accredited Social Health Activists who met with expectant or new mothers to discuss and form strategies to tackle maternal and newborn health issues. Similarly, Townsend utilised the PLA methodology to develop projects to enhance peer-based care for HIV positive prisoners in Malaysia.[25] PLA cycles have also been augmented with interventions such as health systems strengthening[22], cash transfers[27] and home counselling[28]. Harris-Fry et al combined PLA women's group with either cash or food transfers to improve maternal diets and nutrition during pregnancy in the Dhanusha and Mahottari districts in Nepal.[27] Though implementation of PLA improved diets, the addition of cash and food transfers improved the variety of diet and allocation of high energy food to pregnant mothers, respectively.

Similarly, Valdez et al. identified that youth PLA groups had a positive impact on young people in terms of substance use prevention.[24] These interventions mainly took place in the United States, with some studies included from Canada and Bosnia Herzegovina. Benefits included greater awareness of drug and alcohol misuse in their community and decreased approval of peer drug use. This was achieved partly by individual behavioural change as well as increasing youth engagement in social action. Young people were also able to gain skills in new research methods and given the opportunity to interact with key decision-makers within their community.

A phenomenon of 'global innovation', whereby innovations developed, evaluated and implemented in LIC are adapted and tested within high-income countries (HICs), is gaining popularity in HICs.[29] An exemplary initiative would be the Partners in Health's 'Prevention and Access to Care and Treatment (PACT) project. Using the home-based medical and social support service used in rural Haiti, PACT was able to increase medical adherence and therefore, positive medical outcomes for HIV patients in Boston, United States.[30] It has since been adopted across cities in the US and globally. Similarly, The Nurture Early for Optimal Nutrition (NEON) study aims to translate the achievements from Bangladesh, India, Malawi, and Nepal noted in Prost et al's work to improve infant feeding and care practices in Tower Hamlets in London.[31] Further, the co-adaptation of community participatory

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group programmes to support and improve outcomes for parent/carers of children with neurodisability is being co-developed to be ready for piloting in Newham, East London.[32]

A further consideration for the current use of PLA is how the approach is delivered, with whom and for what outcomes and furthermore how to evaluate these approaches within HIC settings. PLA approaches within LICs are established in both online and face-to-face across all economic backgrounds. Face-to-face engagements are the norm for PLA as historically their usage has been within LICs, wherein online infrastructure tends to be less well developed. More recently, online elements have been incorporated into PLA usage in LICs. Unsurprisingly, a higher proportion of PLA applications in HICs implement online components, such as online seminars, forums and surveys for data collection.[9,33] Increased resources and improved data infrastructure enable far more rapid sharing of data and multimodal interventions as a part of PLA, whereas LIC groups cannot utilise these tools to the same extent. Equally, a comparative lack of online resources in LICs results in a predominance of face-to-face PLA groups.

As a proven cost-effective measure with the potential for significant improvement in health outcomes, there is clear applicability of PLA within high-resource settings. Studies to date have variously centred upon the viability and structure of PLA groups in high and low-resource settings, involving multiple populations and contexts.[34,35] At present, no systematic review exists, which synthesises the populations, types of intervention (online or face-to-face) and specific health-related outcomes of PLA groups in HICs.

In this review, we aim to identify the number and types of communities that have engaged in PLA methodology in high-resource settings to date and to analyse any existing documented outcomes of PLA for health and social determinants of health.

METHODS AND ANALYSIS

This protocol was developed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) checklist [36]. This protocol has been registered in PROSPERO (CRD42020187978).

Population, interventions and outcomes

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The review will examine published studies which assess the beneficial effects of PLA groups in social and health outcomes. Interventions can include online or face-to-face groups based in HICs, classified according to the World Bank classification as countries with an income per capita of greater than $12,535.[37] The primary outcome would be any reported improvement related to health or social issues after incorporating the PLA model.

Inclusion and exclusion criteria

We will include randomised controlled trials (RCTs) in HICs that incorporate online or face-to-face group sessions, which apply the learning and action framework. The review will comprise articles published between 2000 and 2020, and exclude any article not written in English.

Aside from participatory action research, we will incorporate studies that incorporate the PDSA and Triple/ Quadruple aim frameworks.

Search strategy

Two reviewers will independently search pre-defined terms in the following electronic bibliographic databases: MEDLINE, EMBASE, CINAHL and the Cochrane Library. Search terms will encompass PLA and PDSA projects, as well as studies using the Triple/ Quadruple Aim model (Table 1).

Table 1 Proposed terms for search strategy

Population	High-income settings filter
Intervention	“action learning" or "active learning" or "action cycle" or "learning cycle" or "experiential learning" or "problem-based learning" or “peer support” or “telephone support” or “home visits” “online support group” or “support group” or “participatory group” or “online participatory group” or “participatory learning and action” or “participatory action” or “community participation” or “PDSA” or “PDCA” or “plan-do-study-act” or “plan do study act” or “plan-do-check-act” or “plan do check act” or “women’s group*” or “quality improv*” or “quality enhanc*”
Outcomes	“triple aim” or “quadruple aim” or “satisfaction” or “patient experience” or “experience of care” or “quality of life” or “health outcome” or “population health” or “per capita” or “cost” or “cost-effectiveness” or “expenditure” or “cost analysis”

Data analysis and synthesis

Abstracts and articles retrieved from the databases will be imported into the EndNote reference management. After removing duplicates, two independent

authors will review the search results to identify those meeting the inclusion criteria. The first stage will screen the studies using only the title and abstracts before proceeding to review the full texts. Our preliminary search indicated that there is little literature available on community groups propagating a PLA cycle to improve specific health and social outcomes in high-income countries. Therefore, we will take care to compare studies with disparate aims. A flow chart will be created to show an overview of the selection process, including the number of excluded papers and the reasons for exclusion (Figure 1). At any stage, if there are any disagreements between the authors, a third reviewer will be involved to resolve it through discussion.

Data will be extracted independently by two reviewers using a standardised, pre-piloted form. The extracted information will include: name of the study, study setting, publication date, journal, authors, sample size, study population, participant demographics and baseline characteristics, population inclusion criteria, details of intervention and control, study methodology, recruitment and study completion rates, length of study, follow-up retention percentage, age of participants, outcomes and times of measurement, indicators of acceptability to users, suggested mechanisms of intervention action, RR, mean differences, type of data analysis, information for risk of bias assessment, source of funding and conflict of interest. Any discrepancies will be resolved through discussion or if necessary, a third reviewer. Missing data will be requested by contacting study authors.

Since we anticipate the outcome of selected studies to vary significantly, we will provide a narrative synthesis of the findings from the included studies without conducting a quantitative analysis. The narrative synthesis will be reported according to the SwiM guidelines and will address the methodology of each study, including assessing for risk of bias.[38] Included studies will be grouped according to the type of intervention, target population characteristics, type of outcome and intervention content.

Due to the expected diverse nature of interventions, we will use a standardised metric to represent the result of each outcome. We will calculate risk ratios for dichotomous outcomes and standardised mean differences for continuous outcomes. The results of each study will be summarised and presented in a tabular form. In doing so, we can systematically assess each study result and assess heterogeneity of key study characteristics. If sufficient data is available, we will conduct a subgroup analysis of different types of participants, for example, women or children.

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Assessment of risk of bias

Two reviewers will independently assess the risk of bias in included studies by considering the Cochrane Collaboration's tool.[39] We will appraise the following characteristics: random sequence generation, allocation concealment, blinding of participants, personnel and outcome assessment, completeness of outcome data, selective outcome reporting and any other potential sources of bias. The reviewers will assess each domain as low, unclear or high risk as per the tool. Any disparities in the assessment of risk between the reviewers will be resolved through discussion, involving a third reviewer where necessary.

DISCUSSION

Following preliminary search, a number of trials exist describing action learning or similar interventions with patient groups in HICs to improve health and social outcomes [40-54]. PLA is a cost-effective and context specific model which has been shown to improve health outcomes in LICs. Therefore, there exists a need to clarify how the field can advance in its understanding of current documented applications of PLA within HIC settings.

We aim to bridge this gap by conducting a systematic review of the current available evidence on PLA in HICs, in order to understand present implementation and better inform future trials. We anticipate a broad heterogeneity in the data gathered owing to the wide contextuality of PLA and the flexibility of recorded outcomes under the PLA philosophy.

Inclusion of both RCTs and trials utilising the PDSA cycle/PLA groups will address direct and indirect usage of PLA and account for all documented use.

We hope that the findings of our review will result in the following outcomes. First, to establish a benchmark in the field for future reference. Second, the ability to elaborate or clarify existing applications of PLA to replicate observed effects. Third, to identify health outcomes that have not previously received attention with regards to implementation of PLA methodology in any setting. Fourth, to examine which PLA modalities relate to which health outcomes and evaluate the significance of these.

We also hope that our review will inform the designing and planning phases of future trials. This could be through guiding possible effect and therefore

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sample sizes, selecting appropriate outcomes and suitable follow-up periods. As it has within several LICs, implementation of PLA could pave the way to influencing healthcare policies and inducing systemic changes to health provision. In summary, a comprehensive review of current literature will enable prioritisation of future research directions.

ETHICS AND DISSEMINATION

No ethics approval is required for this study. The results for this study will be submitted for publication in a leading peer-reviewed academic journal in this field. Additionally, a report will be produced for the funders of this review, which can be viewed for free on their website.

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Authors' statement:

MH, SA, LM, ML conceived the original concept of the study and designed the research methodology. SA, AK, and FM carried out the registration and drafted the manuscript. MH, ML, LM, EW, CS, JM, and AC validated the study and revised the manuscript critically for important intellectual content. SA, AK, FM contributed to the manuscript writing, edited the final manuscript and prepared it for submission. MH, SA had primary responsibility for the final content. All authors read and contributed to reviewing the paper, the designing of the manuscript, and the approval of the final manuscript.

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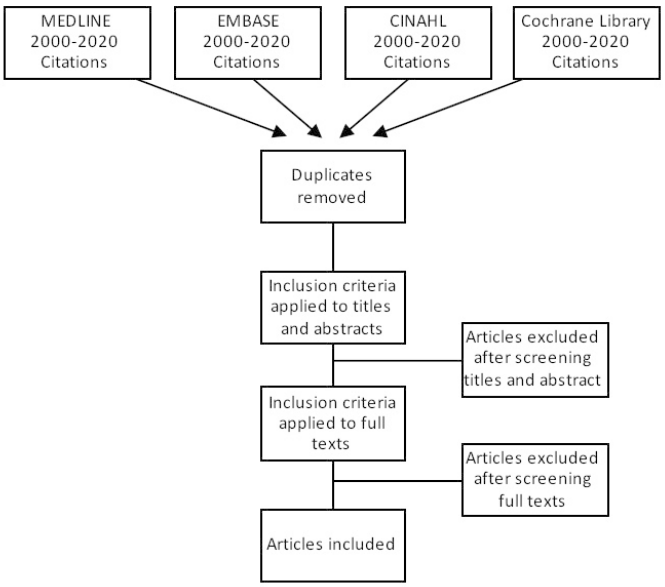
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The data from this study is available from the corresponding author upon reasonable request.

Patient and public involvement statement:
It was not appropriate or possible to involve patients or the public in the design, or conduct, or reporting, or dissemination plans of our research.

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Preferred Reporting Items for Systematic Reviews- flow diagram of the search selection for this systematic review.

Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A - not for an update of a previous systematic review
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1

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Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	15
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A - not an amendment of a previous protocol
Support			
Sources	#5a	Indicate sources of financial or other support for the review	15
Sponsor	#5b	Provide name for the review funder and / or sponsor	15
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	15
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	3-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6,7
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7,8

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Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7,8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7,8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7,8
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6-8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	N/A - quantitative synthesis
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9

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Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

Notes:

- 1b: N/A - not for an update of a previous systematic review
- 4: N/A - not an amendment of a previous protocol
- 15a: N/A - quantitative synthesis The PRISMA-P elaboration and explanation paper is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 28. February 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Participatory learning and action (PLA) to improve health outcomes in high income settings: Systematic Review Protocol

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Participatory learning and action (PLA) to improve health outcomes in high income settings: Systematic Review Protocol

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ABSTRACT

Introduction

Participatory learning and action (PLA) is a form of group reflection and learning with documented efficacy in low-income countries (LICs) to improve social and health outcomes. PLA represents both a learning philosophy and practical framework that could be applied to a variety of contexts. To date, PLA has not been widely implemented within high-income countries (HICs) to improve health and health-related outcomes. We aim to synthesise the literature currently available by means of systematic review, to form a foundation for future applications of PLA methodology in HICs.

Methods and analysis

Two reviewers will independently search pre-defined terms in the following electronic bibliographic databases: MEDLINE, EMBASE, CINAHL and the Cochrane Library. Search terms will encompass PLA and PDSA (Plan, Do, Study Act) projects, as well as studies using the Triple/ Quadruple Aim model. We will include randomised controlled trials that incorporate online or face-to-face components using PLA/PDSA methodology. Our data will be extracted into a standardised pre-piloted form with subsequent narrative review according to Synthesis without meta-analysis (SWiM) guidelines.

Ethics and dissemination

No ethics approval is required for this study. The results for this study will be submitted for publication in a leading peer-reviewed academic journal in this field. Additionally, a report will be produced for the funders of this review, which can be viewed for free on their website.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- Incorporation of multiple participatory research methods to accommodate for the variation in nomenclature describing the intervention
- Broad inclusion criteria to allow for wide heterogeneity of PLA study designs
- Potential low number of RCTs/observational studies for inclusion

- Only includes English language studies from 3 databases

INTRODUCTION

The Participatory learning and action (PLA) approach is one of many methods in the field of participatory research, which also includes Participatory Action Research and Community-Based Participatory Research.[1] In comparison to traditional research methods, the underlying philosophy for these methods is to empower communities to identify their own solutions for issues that affect them, rather than for external stakeholders to enforce a solution on to them.[2,3] The onus for researchers is to seek and include diverse opinions and ensure an open space to discuss these. In doing so, researchers can gain first-hand knowledge from communities, often allowing them to collaborate with marginalised members who had previously low engagement. There are multiple techniques that researchers can use to facilitate this process, including drawing community maps of local resources or landmarks, preference rankings to prioritise issues of the group or Venn diagrams to show the sphere of influence of particular individuals or organisations.[1,3,4]

The PLA group cycle is an iterative process typically led by a facilitator who facilitates the participants through a four-stage cycle of identifying and prioritizing contextual issues, designing strategies to address these issues, putting these strategies into practice and a post-implementation evaluation. It aims to support members to take individual and collective action to solve health problems within their community.[5] Through direct engagement, PLA serves to empower communities to understand the importance of their health and the health of their families and communities, as well as the barriers and facilitators to optimising these health outcomes. Due to its demonstrated efficacy, the World Health Organisation has recommended the implementation of groups practicing PLA in rural low income country settings with reduced access to health services in the context of maternal and newborn health.[6,7]

The current PLA model originated from multiple policies and theories from the last century. In his seminal work 'Pedagogy of the Oppressed', Paulo Freire defined that through critical evaluation, the 'oppressed' could re-evaluate and improve their context.[8,9] Around the late 1970s, Robert Chambers developed a more participatory research tool, 'Rapid Rural Appraisal', as an alternative to more didactic learning methods.[4] This tool later developed into

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the PLA cycle, but similar models have also emerged such as the Plan-Do-Study-Act (PDSA) cycle.[10] The PDSA cycle involves planning an intervention, delivering it and examining its effect before making adjustments based on feedback. As such, it has considerable overlap with the PLA model. However, it should be noted that the initial 'planning' stage of PLA involves stakeholders and therefore a greater element of unpredictability. In contrast, PDSA cycles do not usually have the same extent of collaboration in the planning stage. [11]

Within healthcare settings, PDSA cycles have been incorporated into the 'Model for Improvement' to expedite the Quality Improvement (QI) process.[12] QI is integral to the assessment of health services and delivering better-value, higher quality care for patients, especially important in a climate of rising healthcare costs and limited funding.[13] QI methodology can be applied to review a variety of outcomes, including waiting times, patient experience or post-operative complications. A popular framework for measuring healthcare outcomes is the 'Triple Aim' model.[14] Briefly, the three aims are as follows: improving patient experience of healthcare, improving population health and reducing per capita cost of healthcare. Increasingly, an additional fourth aim, to improve the wellbeing of healthcare workers, is integrated to form the 'Quadruple Aim'.[15]

Thus far, PLA cycles have been successfully incorporated in research to improve health outcomes. Prost et al found that women's groups that utilise PLA initiatives have reduced maternal and neonatal mortality in low-income settings (Nepal[16], India[17–19], Bangladesh[20,21] and Malawi[22,23]).[3] The most significant reduction in neonatal mortality (33%) and maternal mortality (49%) occurred in four of the sites that had the highest coverage of groups and participation of at least 30% of pregnant women. The underlying mechanisms are complex.[1,24] Some benefit can be attributed to effects such as behaviour change for hygiene, nutrition and infant care, as well as social support and birth planning. Additionally, the groups worked towards female empowerment and better decision-making to improve access to basic services, encourage safer delivery practices and reduce delays in reaching care. However, women's groups also provide poor, excluded women a voice by building their confidence to make decisions and to reduce the stigma in their lives, while possibly reducing maternal stress.

The focus of PLA health initiatives in low-income countries (LICs) to date has largely been upon maternal and infant nutrition, but has expanded more recently to address both communicable[25] and non-communicable

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diseases[8,26]. For example, Tripathy et al set up a PLA women's group for pregnant mothers to reduce neonatal mortality in rural Jharkhand and Odisha in Eastern India.[19] Townsend utilised the PLA methodology to develop projects to enhance peer-based care for HIV positive prisoners in Malaysia.[25] PLA cycles have also been augmented with interventions such as health systems strengthening[22], cash transfers[27] and home counselling[28]. Harris-Fry et al combined PLA women's group with either cash or food transfers to improve maternal diets and nutrition during pregnancy in the Dhanusha and Mahottari districts in Nepal.[29]

Other applications of PLA groups are emerging. Valdez et al. identified that youth PLA groups had a positive impact on young people in terms of substance use prevention.[30] These interventions mainly took place in the United States, with some studies included from Canada and Bosnia Herzegovina. Benefits included greater awareness of drug and alcohol misuse in their community and decreased approval of peer drug use.

The wide variety of applications of PLA groups has contributed to a phenomenon of 'global innovation'. Innovations developed, evaluated and implemented in LICs are adapted and tested within high-income countries (HICs), and are gaining popularity in HICs.[29] An exemplary initiative would be the Partners in Health's 'Prevention and Access to Care and Treatment (PACT) project. Using the home-based medical and social support service used in rural Haiti, PACT was able to increase medical adherence and therefore, positive medical outcomes for HIV patients in Boston, United States.[31] It has since been adopted across cities in the US and globally. Similarly, The Nurture Early for Optimal Nutrition (NEON) study aims to translate the achievements from Bangladesh, India, Malawi, and Nepal noted in Prost et al's work to improve infant feeding and care practices in Tower Hamlets in London.[32] Further, the co-adaptation of community participatory group programmes to support and improve outcomes for parent/carers of children with neurodisability is being co-developed to be ready for piloting in Newham, East London.[33]

To understand this phenomenon, it is also important to acknowledge that different patterns of delivery exist between LICs and HICs. Face-to-face engagements are the norm for PLA as historically their usage has been within LICs, wherein online infrastructure tends to be less well developed. Unsurprisingly, a higher proportion of PLA applications in HICs implement online components, such as online seminars, forums and surveys for data collection.[9,34]

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As a proven cost-effective measure with the potential for significant improvement in health outcomes, there is clear applicability of PLA within high-resource settings. Studies to date have variously centred upon the viability and structure of PLA groups in high and low-resource settings, involving multiple populations and contexts.[35,36] At present, no systematic review exists, which synthesises the populations, types of intervention (online or face-to-face) and specific health-related outcomes of PLA groups in HICs.

In this review, we aim to identify the number and types of communities that have engaged in PLA methodology in high-resource settings to date and to analyse any existing documented outcomes of PLA for health and social determinants of health.

METHODS AND ANALYSIS

This protocol was developed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) checklist. This protocol has been registered in PROSPERO (CRD42020187978).

Population, interventions and outcomes

The review will examine published randomised controlled trials (RCTs) and other observational studies in peer-reviewed journals which assess the beneficial effects of PLA groups in social and health outcomes. Interventions can include online or face-to-face groups based in HICs, classified according to the World Bank classification as countries with an income per capita of greater than $12,535.[37] The primary outcome would be any reported improvement related to health or social outcomes after incorporating the PLA model.

Inclusion and exclusion criteria

We will include RCTs and observational studies such as qualitative studies or quality improvement studies in HICs that incorporate online or face-to-face group sessions, which apply the learning and action framework. The review will comprise articles published between 2000 and 2020, to ensure that study interventions are reflective of the current technology and practices. We have decided to exclude any article not written in English, to limit possible risk of misconstruing translated findings from use of online translation services.

Aside from participatory action research, we will incorporate studies that incorporate the PDSA and Triple/ Quadruple aim frameworks.

Search strategy

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Two reviewers will independently search pre-defined terms in the following electronic bibliographic databases: MEDLINE, EMBASE, CINAHL and the Cochrane Library. Search terms will encompass PLA and PDSA projects, as well as studies using the Triple/ Quadruple Aim model (Table 1).

Table 1 Proposed terms for search strategy

Population	High-income settings filter
Intervention	"action learning" or "active learning" or "action cycle" or "learning cycle" or "experiential learning" or "problem-based learning" or "peer support" or "telephone support" or "home visits" "online support group" or "support group" or "participatory group" or "online participatory group" or "participatory learning and action" or "participatory action" or "community participation" or "PDSA" or "PDCA" or "plan-do-study-act" or "plan do study act" or "plan-do-check-act" or "plan do check act" or "women's group" or "quality improv*" or "quality enhanc*"
Outcomes	"triple aim" or "quadruple aim" or "satisfaction" or "patient experience" or "experience of care" or "quality of life" or "health outcome" or "social outcome" or "population health" or "per capita" or "cost" or "cost-effectiveness" or "expenditure" or "cost analysis"

Data analysis and synthesis

Abstracts and articles retrieved from the databases will be imported into the Covidence systematic review management, which automatically detects duplicates and was manually removed.[38] Endnote will be used as reference management software.[39] After removing duplicates using Covidence software, two independent authors will review the search results to identify those meeting the inclusion criteria. The first stage will screen the studies using only the title and abstracts before proceeding to review the full texts. Our preliminary search indicated that the limited literature available on community groups propagating a PLA cycle in high-income countries are to improve specific health or social outcomes. Therefore, we will take care when comparing studies with disparate aims. A flow chart will be created to show an overview of the selection process, including the number of excluded papers and the reasons for exclusion (Figure 1). At any stage, if there are any disagreements between the authors, a third reviewer will be involved to resolve it through discussion.

Figure 1 Proposed Reporting Items for Systematic Reviews flow diagram of the search selection for this systematic review

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Data will be extracted independently by two reviewers using a standardised, pre-piloted form. The extracted information will include: name of the study, study setting, publication date, journal, authors, sample size, study population, participant demographics and baseline characteristics, population inclusion criteria, details of intervention and control, study methodology, recruitment and study completion rates, length of study, follow-up retention percentage, age of participants, outcomes and times of measurement, indicators of acceptability to users, suggested mechanisms of intervention action, RR, mean differences, type of data analysis, information for risk of bias assessment, source of funding and conflict of interest. Any discrepancies will be resolved through discussion or if necessary, a third reviewer. Missing data will be requested by contacting study authors.

Since we anticipate the outcome of selected studies to vary significantly, we will provide a narrative synthesis of the findings from the included studies without conducting a quantitative analysis. The narrative synthesis will be reported according to the SwiM guidelines and will address the methodology of each study, including assessing for risk of bias.[40] Included studies will be grouped according to the type of intervention, target population characteristics, type of outcome and intervention content.

Due to the expected diverse nature of interventions, we will use a standardised metric to represent the result of each outcome. We will calculate risk ratios for dichotomous outcomes and standardised mean differences for continuous outcomes. The results of each study will be summarised and presented in a tabular form. In doing so, we can systematically assess each study result and assess heterogeneity of key study characteristics. If sufficient data is available, we will conduct a subgroup analysis of different types of participants, for example, women or children.

Assessment of risk of bias

Two reviewers will independently assess the risk of bias in included studies by considering the Cochrane Collaboration's tool.[41] We will appraise the following characteristics: random sequence generation, allocation concealment, blinding of participants, personnel and outcome assessment, completeness of outcome data, selective outcome reporting and any other potential sources of bias. The reviewers will assess each domain as low, unclear or high risk as per the tool. Any disparities in the assessment of risk

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between the reviewers will be resolved through discussion, involving a third reviewer where necessary.

Patient and Public Involvement

No patient involved

DISCUSSION

Following a preliminary search, a limited number of studies exist describing action learning or similar interventions with patient groups in HICs to improve health or social outcomes [42-56]. PLA is a cost-effective and context specific model which has been shown to improve health outcomes in LICs. Therefore, there exists a need to clarify how the field can advance in its understanding of current documented applications of PLA within HIC settings.

We aim to bridge this gap by conducting a systematic review of the current available evidence on PLA in HICs, in order to understand present implementation and better inform future trials. We anticipate a broad heterogeneity in the data gathered owing to the wide contextuality of PLA and the flexibility of recorded outcomes under the PLA philosophy.

We hope that the findings of our review will result in the following outcomes. First, to establish a benchmark in the field for future reference. Second, the ability to elaborate or clarify existing applications of PLA to replicate observed effects. Third, to identify health outcomes that have not previously received attention with regards to implementation of PLA methodology in any setting. Fourth, to examine which PLA modalities relate to which health outcomes and evaluate the significance of these.

We also hope that our review will inform the designing and planning phases of future trials. This could be through guiding possible effect and therefore sample sizes, selecting appropriate outcomes and suitable follow-up periods. As it has within several LICs, implementation of PLA could pave the way to influencing healthcare policies and inducing systemic changes to health provision. In summary, a comprehensive review of current literature will enable prioritisation of future research directions.

ETHICS AND DISSEMINATION

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No ethics approval is required for this study. The results for this study will be submitted for publication in a leading peer-reviewed academic journal in this field. Additionally, a report will be produced for the funders of this review, which can be viewed for free on their website.

FIGURE AND TABLE LEGENDS

Table 1 - Proposed terms for search strategy
Figure 1 - Proposed Reporting Items for Systematic Reviews flow diagram of the search selection for this systematic review

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Authors' contributions:

MH, SA, LM, ML conceived the original concept of the study and designed the research methodology. SA, AK, and FM carried out the registration and drafted the manuscript. MH, ML, LM, EW, CS, JM, and AC validated the study and revised the manuscript critically for important intellectual content. SA, AK, FM contributed to the manuscript writing, edited the final manuscript and prepared it for submission. MH, SA had primary responsibility for the final content. All authors read and contributed to reviewing the paper, the designing of the manuscript, and the approval of the final manuscript.

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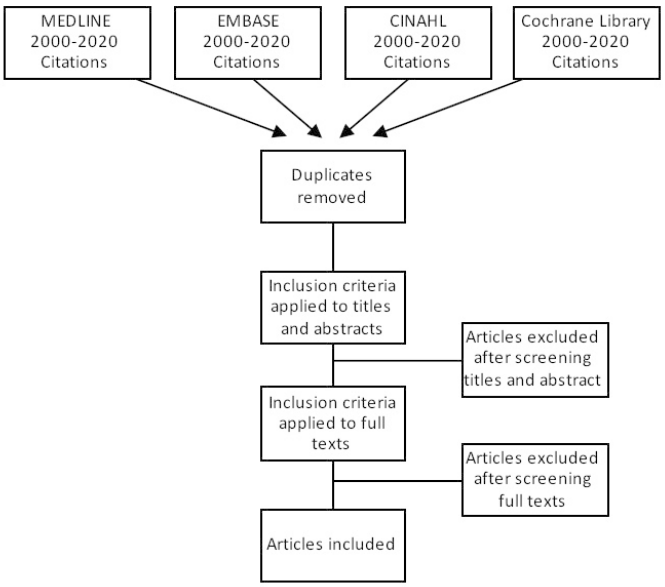
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Competing interests statement.
There is no conflict of interest to declare

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Preferred Reporting Items for Systematic Reviews- flow diagram of the search selection for this systematic review.

Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	1
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A - not for an update of a previous systematic review
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1

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Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	15
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A - not an amendment of a previous protocol
Support			
Sources	#5a	Indicate sources of financial or other support for the review	15
Sponsor	#5b	Provide name for the review funder and / or sponsor	15
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	15
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	3-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6,7
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	7,8

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Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7,8
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7,8
Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7,8
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6-8
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6-8
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	N/A - quantitative synthesis
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	9
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	9

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Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

Notes:

- 1b: N/A - not for an update of a previous systematic review
- 4: N/A - not an amendment of a previous protocol
- 15a: N/A - quantitative synthesis The PRISMA-P elaboration and explanation paper is distributed under the terms of the Creative Commons Attribution License CC-BY. This checklist was completed on 28. February 2021 using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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